BOTTLE DETECTION BUILDER JOURNAL

YOLOv8 Project Development Log

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Week 2 Progress Report | July 2026

This journal documents my journey developing a YOLOv8-based bottle detection system as part of my computer vision coursework. Over the past two weeks, I've encountered various challenges in dataset preparation, model training, and performance optimization. This log captures the key obstacles I faced, solutions I implemented, and valuable lessons learned throughout the development process. The insights gained will guide my approach for Week 3 and future machine learning projects.

📊 Dataset Challenges

Collecting and preparing the bottle detection dataset proved more challenging than anticipated. Initially, I gathered 500 images from various sources, but quickly realized the dataset lacked diversity in lighting conditions, backgrounds, and bottle types. Many images contained cluttered backgrounds that confused the model during early training runs. I spent considerable time curating additional images to include different bottle shapes (plastic, glass, metal), various orientations (upright, tilted, lying down), and diverse environmental conditions (indoor, outdoor, different lighting). The final dataset expanded to 1,200 images with better representation across all categories. This diversity significantly improved the model's generalization capability.

🏷️ Annotation Challenges

Manual annotation of 1,200 images using LabelImg consumed far more time than expected—approximately 15 hours over three days. Maintaining consistency in bounding box placement, especially for partially visible or overlapping bottles, required careful attention and multiple review passes. I initially made errors by drawing boxes too tight around objects, which caused the model to miss bottles with slight variations in appearance. After consulting online resources and reviewing YOLO annotation best practices, I adjusted my approach to include small margins around objects and ensure complete coverage of bottle caps and bases. I also discovered the importance of annotating difficult cases like transparent bottles and bottles in shadows, even though they were harder to label. These challenging examples ultimately helped the model learn robust features.

⚙️ Training Challenges

Training the YOLOv8 model presented several technical hurdles that tested my understanding of deep learning workflows. My first training attempt failed due to incorrect dataset path configuration in the YAML file, which took two hours to debug. Once training began, I noticed the loss values weren't decreasing as expected after 50 epochs, indicating potential issues with learning rate or data augmentation settings. I experimented with different hyperparameters: reducing the initial learning rate from 0.01 to 0.001, increasing batch size from 8 to 16, and enabling additional augmentation techniques like mosaic and mixup. GPU memory limitations on my laptop (6GB VRAM) forced me to use the YOLOv8n (nano) model instead of the larger YOLOv8m variant I originally planned. Despite these constraints, after 100 epochs with optimized settings, the model achieved 87% mAP@0.5, which exceeded my initial target of 80%. The training process taught me the importance of systematic experimentation and careful monitoring of training metrics.

📈 Performance Improvements

Metric	Before Optimization	After Optimization
Precision	0.72	0.89
Recall	0.68	0.85
mAP@0.5	0.71	0.87
Inference Speed	45ms	38ms
False Positives	High	Low

💡 Lessons Learned

- Dataset quality matters more than quantity—diverse, well-annotated images outperform large homogeneous datasets
- Consistent annotation standards are critical; I now use a checklist to ensure uniform bounding box placement
- Starting with a smaller model (YOLOv8n) allowed faster iteration and experimentation within hardware constraints
- Monitoring training curves in real-time helps identify issues early; I learned to recognize overfitting patterns
- Data augmentation techniques like mosaic and rotation significantly improved model robustness to new scenarios
- Patience is essential—rushing through annotation or training leads to poor results that require rework
- Documentation throughout the process saves time when troubleshooting or explaining methodology

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🎯 Week 3 Goals

- [ ] Expand dataset to 1,500 images with focus on edge cases (transparent bottles, extreme lighting)
- [ ] Implement real-time detection using webcam feed for live testing and demonstration
- [ ] Optimize model for deployment on edge devices (Raspberry Pi or Jetson Nano)
- [ ] Create comprehensive documentation including architecture diagrams and API specifications
- [ ] Conduct comparative analysis with other detection models (Faster R-CNN, SSD)
- [ ] Prepare final presentation with demo video and performance benchmarks

📅 Project Timeline

Week 1	Dataset collection and annotation (500 images)
Week 2	Model training, optimization, and performance tuning
Week 3	Real-time implementation and documentation
Week 4	Final testing, presentation preparation, and project delivery